



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

CV Assist

*A Real-Time Visual Assistance System
for the Visually Impaired*

Key Technologies:

- Open-Vocabulary Detection (OWL-ViT v2)
- Monocular Depth Estimation (MiDaS)
- Hand Tracking (MediaPipe)
- LLM-Vision Error Correction

Developer: CV Assist Team

CONTENTS

- 1 Project Motivation & Challenges
- 2 System Architecture Design
- 3 Core Perception Technologies
- 4 Interaction & Stability Optimization
- 5 Conclusion & Future Work



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

01

PART ONE

Project Motivation & Challenges

1. Motivation: Breaking Barriers

"Existing tools lack real-time, intuitive spatial guidance for daily object interaction."

Blurred Perspective (Current)

CV Assist Guided (Goal)

**Core Pain Point:
Independent Grasping**

Design Goals & Vision

Real-Time

Millisecond visual feedback & voice commands

Versatility

Support for any object via open-vocabulary search

Intuitiveness

Natural language guidance for precise touch

Motivation

Architecture

Technologies

Conclusion

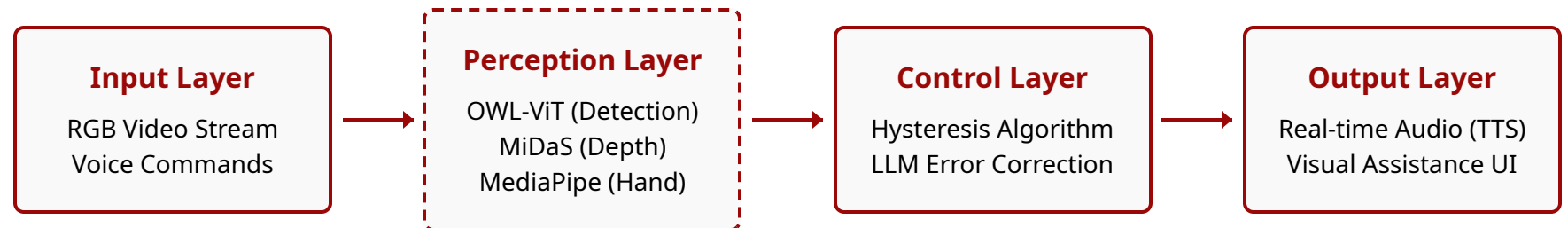
2. Architecture: Modular Pipeline Design

Architecture

Implementation

Technologies

Conclusion



- **Decoupled Design:** Modular pipeline ensures independent optimization of each detector.
- **Multimodal Fusion:** Combining visual depth and voice semantics for robust guidance.



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

02

PART TWO

Core Perception Technologies



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

Perception: OWL-ViT & MiDaS



OWL-ViT v2

Open-Vocabulary Detection



MiDaS_small

Monocular Depth Estimation

Zero-Shot Capability:

- Identify any object without pre-training
- Support natural language search commands
- Robust performance in complex backgrounds

Spatial Awareness:

- Real-time depth map from a single lens
- Estimate relative distance to hand/target
- Low-power design for mobile deployment

Motivation

Architecture

Core Tech
Implementation

Conclusion

04 Interaction: MediaPipe & LLM Error Correction

MediaPipe Hand Tracking

- **High-Precision Tracking:** Real-time 21 3D hand landmarks for spatial alignment.
- **Gesture Recognition:** Detects "Open" and "Closed" states for user intent.
- **Centimeter Precision:** Fine-grained navigation via real-time distance calculation.

LLM-Vision Interaction

- **Multimodal Fusion:** Combines Whisper ASR text with recent 4 video frames.
- **Semantic Alignment:** DeepSeek-v3.2 corrects ASR errors (e.g., "cup" vs "cap").
- **Robustness:** Maintains interaction accuracy in noisy environments via visual context.

"By fusing visual perception with linguistic intelligence, the system evolves from 'seeing objects' to 'understanding intent,' providing a natural and reliable user experience."



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

03

PART THREE

Conclusion & Future Work

Thanks for listening

CV Assist: Empowering the Future of the Visually Impaired

Developer: CV Assist Team
